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Affective empathy mediates the positive effect of prosocial video games on young children's sharing behavior

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ABSTRACT

This study investigated the effect of prosocial video game exposure on sharing behavior in Chinese preschoolers and tested the mediating role of empathy as well as the moderating roles of age and gender. A total of $N = 120$ children aged 4–6 years (50% girls, 50% boys; $M_{age} = 5.0$, $SD = 0.82$) from two Chinese kindergartens participated in this between-subjects design experiment. The results showed that the preschoolers who played prosocial video games exhibited more sharing behavior than those who played neutral video games. Empathy, especially affective empathy, was found to mediate the association between playing prosocial video games and sharing behavior. This suggests that empathy training programs, especially those that promote affective empathy, may be an effective means of promoting sharing behavior in children. The mediating effect was mainly observed for girls and older children. Girls and older children may therefore be the target group for sharing behavior education programs that aim to increase empathy in playing prosocial video games. Nevertheless, promoting sharing behavior in younger children and boys is equally important and deserves attention in future research.

1. Introduction

Prosocial behavior is defined as behavior through which people benefit others (Ding et al., 2018; Eisenberg & Miller, 1990), and there are three distinct types (helping, sharing, and comforting) within the general domain of prosocial behaviors (Geraci & Franchin, 2021). As a typical prosocial behavior, sharing behavior refers to a willingness to sacrifice personal gain for the needs of others (Ongley & Malti, 2014). Children are more likely to have good peer relationships and better social outcomes if they are more willing to share with others (Samek et al., 2020). This skill is particularly important for 3– to 6-year-old children who may be less likely to share valuable resources at their own expense (Blake & Rand, 2010; Fehr et al., 2008). However, there is a paucity of research on how to improve sharing behavior in preschool children.

In recent years, the prominence of video games has grown significantly, becoming an increasingly integral aspect of the lives of children and adolescents (Domahidi et al., 2014; Shoshani & Krauskopf, 2021). Global statistics indicate that there are approximately 3.09 billion active video gamers worldwide, and this number is expected to increase to 3.32 billion by 2024 (Howarth, 2023). Even among young children aged 3–6 in urban areas of China, there is a notable prevalence of avid gamers (Shan et al., 2010; Sun et al., 2014). The widespread popularity of video games among children provides a clear opportunity to improve their sharing behavior through prosocial video games. A prosocial video game (PVG) is a game with prosocial content, such as helping, as a theme (Chambers

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& Ascione, 1987; Rosetti et al., 2020). Although several studies have examined the relationship between playing PVGs and other types of prosocial behavior such as helping (Greitemeyer & Osswald, 2010; Prot et al., 2014; Saleem et al., 2012), collaboration (Gentile. et al., 2009), and charitable giving (Colleges & Tom, 2019), there is little research on whether PVGs promote sharing behavior in preschool children and the underlying mechanisms.

1.1. Social learning theory

Social learning theory is commonly used to explain children's social learning process, which states that individuals can acquire new behaviors by observing the behavior of others and the consequences that follow (Bandura, 1969b; Bandura et al., 1963). Such observation can take place in a variety of settings, including physical spaces, television, and films. The way an individual respond to the observed behavior depends on the input variables, such as situational (e.g., types of video games, level of game immersion, provocation) and personal (e.g., traits, gender, age) factors (Buckley & Anderson, 2006; Zhang, Cao, & Tian, 2021b). These factors can affect one's internal state, which comprises cognition, affect, and arousal, ultimately leading to a mediating process that determines whether the behavior is imitated or not (Bandura, 1969a). Additionally, the relationship between video game play and children's social behavior could also be explained by the general learning model, which suggests that behavioral scripts acquired through video game play can guide players' behavior in similar real-world situations (Sarmet & Pilati, 2016). Drawing from social learning theory and other related theories, it is possible that playing PVGs could have a positive effect on children's sharing behavior. This effect would likely be mediated by the internal state of the child, which may be influenced by the children's personal and situational factors.

Bandura's social learning theory was developed prior to the proliferation of digital media, thus, there is a need to extend its application to contemporary contexts. One promising area for such extension is the study of the effects of video games on social learning. By examining how PVGs affect the sharing behavior of young children, this study contributes to the advancement of Bandura's theory in new domains of application.

1.2. Prosocial video games and sharing behavior

In contrast to violent video games that simulate aggression and violence, PVGs aim to promote cooperation, empathy, and altruism (Anderson et al., 2010; Harrington & O'Connell, 2016). Research has shown that playing PVGs can enhance children's prosocial tendencies and behavior (Anderson et al., 2010; Harrington & O'Connell, 2016). For instance, one PVG called *The Empathy World* is an interactive video game that exposes players to diverse prosocial situations (e.g., crying baby), promoting empathy and understanding among players (Wu & Kim, 2019). Recent studies have also suggested that playing PVGs can positively influence children's socio-emotional functioning (Shoshani & Krauskopf, 2021), peer interactions (Verheijen et al., 2019), and helping behaviors (Wiederhold, 2021).

Sharing, as a fundamental social behavior, is directly relevant to the development of children's positive peer relationships and overall social growth, particularly in preschool children (Blake & Rand, 2010; Samek et al., 2020). The relationship between giver and receiver plays a crucial role in resource allocation in young children. Specifically, children are more likely to share with familiar individuals, such as close friends or family members, than with unfamiliar individuals (Fehr et al., 2008). Moreover, children exhibit a preference for sharing with good friends as opposed to neutral familiar peers (Engelmann et al., 2021). This preference can be attributed to a sense of reciprocity and trust that children associate with familiar individuals and close friends, alongside an expectation of positive outcomes resulting from interactions with these individuals (Tomasello et al., 2005).

Despite the crucial role that sharing plays in children's social development, few studies have examined the potential positive effects of video games on sharing behavior in the current context of widespread video game use. Gentile and colleagues (2009) used the *Prosocial Orientation Questionnaire* (e.g., "When I see a student who is upset, it really bothers me.") to examine the relationship between PVG use and prosocial behavior among adolescents. Participants were asked to give their answers on a Likert scale. The findings indicated that a higher frequency of PVG use among adolescents was associated with increased engagement in helping, cooperating, and sharing behaviors. Shoshani and colleagues suggested that involvement in PVGs provides adolescents with increased opportunities to practice and refine their prosocial skills, particularly in the area of sharing among individuals aged 9–15 years (Shoshani et al., 2021). They employed the questionnaire *Strengths and Difficulties Questionnaire* (e.g., "I usually share with others, for example CD's, games, food.") and participants reported their responses using a Likert scale ranging from "not true" to "certainly true." Additionally, Harrington and O'Connell (2016) found a positive association between PVG use and the performance of prosocial behaviors (including sharing) in children and adolescents aged 9–15 years. They used the *Prosocial Orientation Questionnaire* (e.g., "I feel jealous when my friends win an award or prize.") to measure participants' prosocial tendencies, and children had to respond using a Likert scale. The available evidence from questionnaire studies suggests that playing a PVG may have a positive effect on children's sharing behavior.

1.3. The mediating role of empathy

Although playing PVGs may promote sharing behavior in individuals, the underlying mechanisms behind this relationship remain unclear. One possible explanation is the mediating role of empathy, which has been found to be positively associated with both prosocial behavior and PVG exposure (Decety et al., 2016; Zhang & Wang, 2019). In the following section, the present work will take a closer look at empathy and its mediating role in the association between PVGs and sharing behavior.

Empathy reflects the ability to perceive and be sensitive to the emotional states of others, often coupled with a motivation to care for the well-being of others (Decety & Michalska, 2020). Trait empathy, on the other hand, represents an enduring characteristic that

indicates an individual's consistent inclination to experience empathy across various contexts and over time (Barrio et al., 2004). The structure of trait empathy includes both affective and cognitive elements (Winters et al., 2021). Affective empathy (AE) involves emotional resonance with others while recognizing that their feelings are distinct from one's own (Mazza et al., 2014; Schwenck et al., 2012). For instance, even newborns respond to the cries of other infants of the same age, demonstrating early forms of AE (Geangu et al., 2011). Children's AE develops gradually through reciprocal interactions with caregivers (Decety & Holvoet, 2021). Cognitive empathy (CE) refers to adopting others' views and attributing their thoughts and feelings (Decety & Cowell, 2015). It develops somewhat later than AE (Bensalah et al., 2016). The cognitive component of empathy matures when children can take another person's perspective, reflect on their emotions, and offer solutions or help when they notice someone in distress (Decety & Holvoet, 2021).

Based on the social learning theory, there is a mediating process between observing the behavior (stimulus) and imitating it or not (response) (Bandura, 1969a). Previous research has shown that the relationship between PVG use and prosocial behavior was mediated by the role of empathy (Gentile et al., 2009; Gentile et al., 2014; Zhang & Wang, 2019). Specifically, a longitudinal study found that PVG use was a significant predictor of helping and cooperation in children and adolescents, and that this change was mediated by empathy (Prot et al., 2014). In this study, empathy was measured by the Empathic Concern and Perspective Taking subscales of the Interpersonal Reactivity Index which is designed to measure empathy as a set of constructs in addition to a unitary concept. For instance, "Before criticizing somebody, I try to imagine how I would feel if I were in their place." Participants are required to give responses based on the Likert scale. Another study also highlighted that the mediating effect of empathy concern exists between exposure to prosocial television content and prosocial behavior in adolescents. Participants' empathy concern was measured by seven items such as "When I see someone being taken advantage of, I feel kind of protective towards them", which is reported by children using the Likert scale (Padilla-Walker et al., 2015). As a result, it is reasonable to expect that empathy may act as a mediator between PVG exposure and sharing behavior.

1.4. Gender and age as moderators

Although previous literature suggests that empathy might mediate the effect of playing PVG on prosocial behavior, the magnitude of this effect may vary depending on personal characteristics such as age and gender (Buckley & Anderson, 2006). First, age may act as a potential moderator of the mediation process, as studies have shown that empathic responses to stories increase with age, with older children displaying greater empathy (Lennon et al., 1986; Marcus et al,., 1979). Similarly, sharing behavior also increases with age (Li et al., 2016; Liu et al., 2016; Malti et al., 2016). Nine-year-olds, for example, shared significantly more stickers with anonymous classmates than four-year-olds (Benenson et al., 2007). Second, the development of prosocial behavior differs between boys and girls (Baillargeon et al., 2011). Girls display more sharing behavior than boys in general (Baillargeon et al., 2011; Burford et al., 1996) and they also exhibit greater empathy than boys in response to adverse events that happen to others (Benenson et al., 2021; Li et al., 2018). In this vein, it is possible that girls and older children may have an advantage over boys and younger children in gaining higher levels of empathy from playing PVG and, therefore, exhibiting higher levels of sharing behavior.

1.5. The present study

The existing research literature has several shortcomings that need to be addressed. Firstly, some researchers have focused on other types of prosocial behavior, such as helping, collaborating, and giving behavior (Colleges & Tom, 2019; Gentile, 2009; Greitemeyer & Osswald, 2010; Prot et al., 2014; Saleem et al., 2012), and there is little direct empirical evidence on how to improve children's sharing behavior in the context of playing PVGs. Secondly, although studies have demonstrated the potential mediating effect of empathy between playing PVGs and prosocial behavior in elementary school children (Gentile. et al., 2009) and adolescents (Prot et al., 2014), few studies have investigated the potential mechanisms of empathy in preschool children. Furthermore, previous studies have mainly relied on questionnaires, and there is a need for more experimental research. The present study aims to provide experimental evidence on the mediating role of empathy in the development of sharing behavior in children playing video games. Thirdly, it is still unclear whether the cognitive and/or affective pathways mediate the effect of video game exposure on action (Greitemeyer & Osswald, 2010, 2011), which is crucial when choosing the type of video games for children. Additionally, most studies on this topic have been

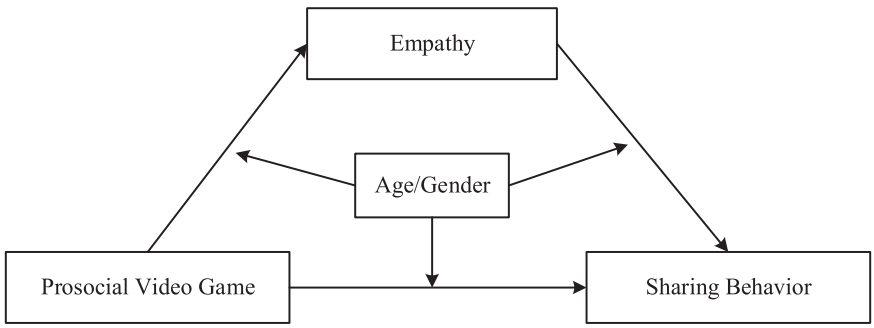

Fig. 1. *Proposed Moderated Mediation Model. Note.* 0 = playing neutral video game, 1 = playing PVG.

conducted in a Western context, and it is essential to test the effects of video games in non-Western samples, as children from different cultures may have different social preferences and behaviors (Póvoa et al., 2020). Thus, this study can provide new insights into the effects of playing video games on sharing behavior and offer new perspectives on how to develop sharing behavior in kindergarten children.

The aim of the present study was to investigate the relationship between playing PVGs and children's sharing behavior. In particular, the present work aims to analyze the role of empathy as a mediator in this relationship, as well as the moderator roles of gender and age in this mediation effect (Fig. 1). The following research questions guided this study: (1) Does playing a PVG increase children's sharing behavior? (2) If so, does empathy mediate the effects of the PVG on sharing behavior? (3) Is such a mediating effect different for different age groups of children? (4) Is this mediating effect different for boys and girls? Based on the previous literature and the primary research questions, the present work proposes the following research hypotheses.

Hypothesis 1. Playing a PVG will increase sharing behavior in preschool children. It has been shown that playing a PVG positively predicts children's prosocial behavior, such as helping (Greitemeyer & Osswald, 2010), extending the findings to sharing behavior. Hypothesis 2: The effect of PVG on sharing will be mediated by empathy. According to social learning theory, there may be a mediating mechanism that links the observation of behavior with the subsequent decision to imitate or not (Bandura, 1969a). Empathy plays a crucial role in mediating the relationship between exposure to prosocial media and the manifestation of prosocial behavior (Greitemeyer, 2009; Greitemeyer et al., 2010; Prot et al., 2014). Hypothesis 3: The mediating effect of empathy between PVG play and sharing behavior will be more significant for older than younger children. As children mature, they tend to show more prosocial behavior and their empathy is more easily activated with age (Brink et al., 2011; Li et al., 2016), which means that they are more likely to benefit from PVG play. Hypothesis 4: The mediating effect of empathy between playing PVG and sharing behavior will be more significant for girls than for boys. Girls generally experience more empathy than boys for others' plights (Busching & Krahé, 2020).

2. Method

2.1. Participants

The present work collected the data from 120 preschoolers aged 4–6 years (50% girls; 50% boys; *M*age = 5.0, *SD* = 0.82) from two kindergartens in Chongqing, China. Sixty preschoolers (50% girls; 50% boys; $N_{4\text{-year-olds}} = 20$, $N_{5\text{-year-olds}} = 20$, $N_{6\text{-year-olds}} = 20$) were randomly assigned to play a PVG, while the other 60 preschoolers (50% girls; $N_{4\text{-year-olds}} = 20$, $N_{5\text{-year-olds}} = 20$, $N_{6\text{-year-olds}} = 20$) played a neutral video game. None of the participants had an intellectual disability. The Northeast Normal University Ethics Committee approved the experiment, and the present work obtained written and informed consent from all participants and their parents.

2.2. Materials

2.2.1. Prosocial video games

The prosocial and neutral video games were selected from six video games. The present work conducted a pilot study in which 120 adults (40 parents, 40 kindergarten teachers, and 40 child psychologists) rated the six video games regarding enjoyment, difficulty, prosocial content, prosocial scene, engagement, and action to exclude potential confounding variables. As shown in Table 1, there are significant differences between the six games for prosocial content, F (5, 114) = 4.85, $p < .001$, $\eta^2 = 0.18$, and prosocial scenes, F (5, 114) = 4.04, $p = .002$, $\eta^2 = 0.15$. A post hoc Bonferroni test was conducted to determine which specific groups differed from each other. Results indicated that "*Diego Tractor*" scored a significantly higher score than "*Zuma*", $p = .02$, 90% CI [0.11, 2.40], and "*Tetris*", $p = .001$, 90% CI [0.46, 2.74], in prosocial content. Similarly, "*Diego Tractor*" had a higher score than "*Zuma*", $p = .04$, 90% CI [0.04, 2.16], and "*Tetris*", $p = .02$, 90% CI [0.09, 2.21], in prosocial scenes. Although "*Zuma*" was also significantly different from "*Diego Tractor*", "*Tetris*" was rated even lower in the prosocial content and scenes. Therefore, in order to maximize the differences in the levels of prosocial content and scenes, the present study chose '*Diego Tractor*' as a PVG and '*Tetris*' as a neutral video game.

"*Diego Tractor*" is an action-click game in which players must help Diego to clean up the rubbish and waste on the streets of his community and take it to the dump. Prior to embarking on the gaming experience, the experimenter informed the children about the

Table 1
Means, Standard Deviation, and ANOVA of Enjoyment, Difficulty, Prosocial Content, Prosocial Scene, Engagement and Action.

Measure	DT		CR		SP		FV		Zuma		Tetris		F (5,114)	η2
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*		
Enjoyment	4.75	0.55	4.35	0.67	4.45	0.61	4.20	0.70	4.20	0.70	4.45	0.51	2.15	0.09
Difficulty	1.95	0.6	1.85	0.88	1.85	0.75	1.7	0.66	1.95	1.00	1.90	0.79	0.27	0.01
Content	3.85	1.23	3.40	1.27	3.40	1.14	2.90	1.25	2.60	1.23	2.25	1.07	4.85 * **	1.18
Scenes	3.55	1.05	3.45	0.95	3.20	1.06	2.75	1.29	2.45	1.19	2.40	1.14	4.04 * *	0.15
Engagement	4.20	0.62	4.10	0.85	4.35	0.67	3.95	0.89	4.35	0.67	4.00	0.80	1.03	0.04
Action	4.55	0.51	4.35	0.59	4.35	0.67	4.50	0.51	4.55	0.51	4.30	0.66	0.76	0.03

Note. Number of participants = 120. *ANOVA* = analysis of variance; DT = Diego Tractor; CR = Chicken Run; SP=Save the Puppies; FV = Fruits and Vegetables.
* * $p < .01$; * ** $p < 0.001$.

accumulated rubbish in Diego's community and encouraged them to assist in its cleanup. After the player has moved the rubbish to the dump successfully, a screen appears with subtitles such as "Perfect". "*Diego Tractor*" could promote children's social development through cooperative play and empathy building, as they have to consider the needs and perspectives of others while playing the game. "*Tetris*" requires players to move, rotate and place all sorts of automatically generated blocks to make a complete line and earn points. Similarly, children receive commendations such as "good job" as they progress through the game.

2.2.2. Empathy

The study used the Basic Empathy Scale (BES) (Jolliffe & Farrington, 2006) to measure children's self-reported empathy immediately following their participation in the PVG. In light of previous research on trait empathy that examined the mediating role of empathy in the relationship between prosocial media use and the development of prosocial behavior in children (Prot et al., 2014), the present study also measured children's trait empathy. Although the BES was originally developed to assess empathy in adolescents, some scholars claim that it can also effectively assess empathy in young children (Noelia et al., 2014). The BES consists of two distinct subscales, AE and CE. The AE scale consists of 11 items (e.g., "if your friend feels sad today, you will feel sad, too."), and 9 items measure the CE (e.g., "you can easily realize that your friends are happy today."). Each item asks participants to respond on a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*). The sum of the two subscales is calculated as an overall measure of empathy. The total measure of BES showed accepted internal consistency (Cronbach $\alpha = .67$), as did the CE and AE subscales (Cronbach $\alpha = .79$; Cronbach $\alpha = .85$).

2.2.3. Sharing behavior

Sharing behavior was measured using the dictator game which is a simple decision-making task in which players divide the number of resources between themselves and the recipient (Kahneman et al., 1986; Zhang & Wang, 2019). In order to mitigate the influence of social desirability bias, ten kindergarten teachers and ten children were interviewed to ascertain children's preferences in terms of material possessions. The popularity ratings for stickers ($M = 4.40$), toy cars ($M = 3.20$), plasticine ($M = 3.20$), and Barbie ($M = 3.10$) were obtained from pilot participants on a 5-point scale. Stickers were subsequently chosen as the sharing material for the current investigation. It is worth noting that stickers have been frequently employed as sharing materials in previous studies examining children's sharing behaviors (Kanngiesser & Warneken, 2012; Smith et al., 2013). In addition, to increase the ecological validity of the sharing task, the sharing subject was designated as the children's good friend to create a more realistic sharing scenario, as children are often faced with sharing situations with their good friends (Birch & Billman, 1986). In the study, the experimenters asked each participant the following question: "You have done a good job playing video games, and we are giving you five stickers as a reward. But how many would you share/give to your good friend? You can give none or all to the other child, and it does not matter if you do not want to share with them." Zero stickers indicate no sharing behavior, while one sticker to five stickers represents one point to five points respectively. Higher scores, therefore, indicate a higher level of sharing behavior, and lower scores indicate a lower level of sharing behavior.

2.3. Procedure

The experiment was conducted in a quiet room in each kindergarten. Before the experiment, experimenters explained and assured the participants that they were free to decline participation without negative consequences. In line with previous experimental research on the effects of video games on children, participants were allowed to engage in video game activities for a period of 15 min (Zhang et al., 2021a). The present work followed a similar methodology in which children were randomly assigned to play either the PVG or the neutral video game for a period of 15 min. The participant then answered the BES questions verbally with the help of the experimenters. After that, participants completed the sharing behavior assessment. The trained assistants explained all procedures, answered questions, thanked the participant for his or her cooperation, and gave him or her stickers.

2.4. Data Analysis

The present work conducted all of the analyses in R. First, the present work performed independent samples *t*-test to investigate whether there were different levels of sharing between the PVG group and the control group. Then, the present work applied mediation models (Chen, Aryee, & Lee, 2005) using the 'mediation' package (Sales, 2017) to investigate the mediating influence of empathy. Finally, the present work examined the moderating influence of age and gender using moderated mediation analysis as outlined in previous research (Preacher, Rucker, & Hayes, 2007). In both the mediation and moderated mediation models, the present work used bootstrapping (Preacher & Hayes, 2004) to provide reliable standard errors for parameter estimation. Power analyses in G*power with an α-level of .05 indicated adequate statistical power for *t*-test and mediation analysis $(1 - \beta) > .80$.

3. Results

3.1. Differences between neutral and prosocial video game groups

Table 2 shows the means, and standard deviations of empathy and sharing behavior in the 12 subgroups. The *t*-test results revealed that the group of participants who played the PVG ($M = 4.00$, $SD = 1.22$) showed a statistically significant increase in sharing behavior compared to the group who played the neutral video game ($M = 3.08$, $SD = 1.44$), $t\ (1, 18) = -3.76$, $p = .000$, $d = 0.69$ (Fig. 2).

Table 2
Means and Standard Deviations of Empathy and Sharing Behavior.

Games	Age	Empathy				Sharing Behavior			
		Boys (50%)		Girls (50%)		Boys (50%)		Girls (50%)	
		M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Prosocial	4-year-old	3.32	0.53	3.81	0.38	2.90	1.85	3.90	0.88
	5-year-old	3.66	0.64	3.99	0.39	4.00	1.25	4.60	0.52
	6-year-old	4.12	0.42	4.22	0.40	3.80	1.03	4.80	0.42
Neutral	4-year-old	3.64	0.42	3.42	0.57	3.60	1.35	3.30	1.57
	5-year-old	3.86	0.49	3.67	0.59	3.60	1.78	2.10	0.99
	6-year-old	3.49	0.51	3.73	0.64	2.20	0.79	3.70	1.25
Total		3.68	0.55	3.80	0.55	3.35	1.47	3.73	1.33

Note. Number of participants = 120.

3.2. The mediating effect of empathy

To explore the relationships between key variables, the present study performed regression analyses, and the findings are summarized in Table 3. The results suggested that empathy could potentially act as a mediator in the association between playing PVG and sharing behavior. Based on the results of mediation model, the mediated path from PVG through empathy to sharing behavior was statistically significant (ACME[1] = 0.18, 95% CI [0.02, 0.46], $p = .032$) and the proportion was approximately 17.90%. The PVG significantly predicted empathy (effect = 0.22, 95% CI [0.03, 0.37], $p = .024$), and sharing behavior (effect = 0.75, 95% CI [0.30, 1.20], $p = .001$). Empathy could also significantly predict sharing behavior (effect = 0.75, 95% CI [0.27, 1.21], $p = .001$).

The present work also tested whether AE and CE respectively mediated the effects of PVG on sharing behavior. As shown in Fig. 3, the mediating effect of PVG on sharing behavior through AE was significant (ACME = 0.15, 95% CI [0.01, 0.45], $p = .030$) and the proportion was about 15.09%. PVG significantly predicted AE (effect = 0.27, 95% CI [0.03, 0.52], $p = .031$), while AE significantly predicted sharing behavior (effect = 0.50, 95% CI [0.15, 0.85], $p = .005$). However, the mediated path from PVG via CE to sharing behavior was not significant (ACME = 0.07, 95% CI [−0.03, 0.27], $p = .210$). PVG did not significantly predict CE (effect = 0.15, 95% CI [−0.07, 0.38], $p = .181$), whereas CE significantly predicted sharing behavior (effect = 0.43, 95% CI [0.04, 0.82], $p = .030$).

In conclusion, the present work suggests that empathy serves as a mediator in the relationship between playing PVG and children's sharing behavior. Specifically, AE rather than CE mediates the effect of PVG on sharing behavior.

3.3. Moderated mediation of age

The present work tested the potential moderating effects of age and gender on the application of mediation models. First, the present study examined the role of age as a moderator in the mediation model (Fig. 4). The results showed that age moderated the effect of playing the PVG on children's empathy performance (effect = 0.26, 95% CI [0.03, 0.49], $p = .024$), but not on their sharing behavior (effect = 0.42, 95% CI [−0.17, 1.02], $p = .163$). However, age did not moderate the effect of empathy on children's sharing behavior (effect = 0.34, 95% CI [−0.21, 0.90], $p = .220$). The present work also found that the mediating effect was significant for children aged 5 years (ACME = 0.15, 95% CI [0.04, 0.34], $p = .000$) and 6 years (ACME = 0.49, 95% CI [0.16, 0.56], $p = .000$), but not for children aged 4 years (ACME = −0.01, 95% CI [−0.19, 0.03], $p = .800$). The proportion of mediating effects also increased from 17.06% for 5-year-olds to 30.1% for 6-year-olds.

The present work also conducted further tests to examine the moderating effect of age on the mediating role of AE. The results indicated that age moderated the direct effect of playing the PVG on AE (effect = 0.39, 95% CI [0.10, 0.69], $p = .010$) as well as the mediating effect of AE. More precisely, age moderated the mediating effect of AE between playing PVG and sharing behavior for children aged 5 years (ACME = 0.11, 95% CI [0.04, 0.22], $p = .000$) and 6 years (ACME = 0.47, 95% CI [0.22, 0.65], $p = .000$) with the proportion of 13.41% and 28.60%, respectively. However, the mediating effect of AE was not significant for 4-year-olds (ACME = −0.01, 95% CI [−0.15, 0.00], $p = .400$).

Overall, age moderates the mediating effect of empathy between playing PVG and sharing behavior, with a stronger effect observed for older children. Moreover, the proportion of mediating effects of empathy increases with increasing age. This trend was also observed for the mediating effect of AE.

3.4. Moderated mediation of gender

The results showed that although gender did not moderate the direct effect of playing PVG on empathy (effect = 0.30, 95% CI [% CI [−0.01, 0.74], $p = .053$) or prosocial behavior (effect = 0.63, 95% CI [−0.32, 1.59], $p = .193$), the mediating effect of empathy differed between girls and boys. More concretely, the mediating effect of empathy was significant for girls (ACME = 0.35, 95% CI [0.16, 0.50], $p = .000$) with a proportion of 25.14%, but not for boys (ACME = 0.02, 95% CI [−0.03, 0.12], $p = .400$). Likewise, the

[1] ACME stands for average causal mediation effects.

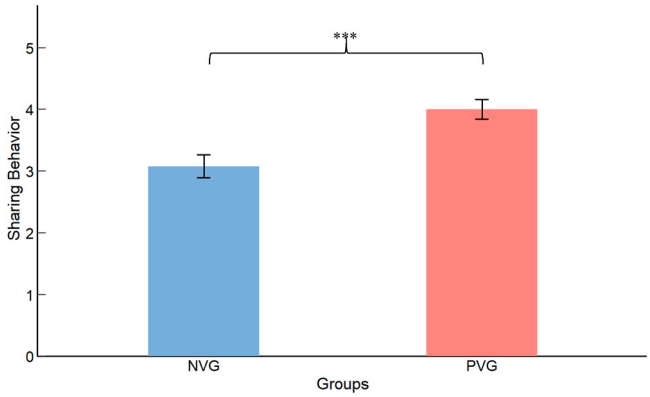

Fig. 2. *Group Differences in Sharing Behavior. Note.* PVG, prosocial video game; NVG, neutral video game. Error bars represent standard errors. * $p < .05$; * * $p < .01$; * ** $p < 0.001$.

Table 3
Results of Regression Analysis of Empathy and Sharing Behavior.

Effect	Estimate		*SE*	95% CI		*p*
		β		*LL*	*UL*	
Empathy						
Intercept	2.586		.333	1.927	3.244	.000
Gender[a]	.126	.115	.096	-.063	.315	.190
Age	.171	.200	.059	.055	.287	.004
Games[b]	.219	.255	.096	.030	.408	.024
Sharing behavior						
Intercept	.072		1.001	-1.911	2.055	.942
Gender[a]	.289	.103	.234	-.176	.754	.221
Age	-.028	-.016	.148	-.321	0.265	.849
Games[b]	.753	.268	.238	.280	1.225	.002
Empathy	.749	.293	.226	.300	1.197	.001

Note. Number of participants = 120. CI = confidence interval; *LL* = lower limit; *UL* = upper limit.
[a] 1 = boys, 2 = girls. [b] 0 = playing neutral video game, 1 = playing prosocial video game.

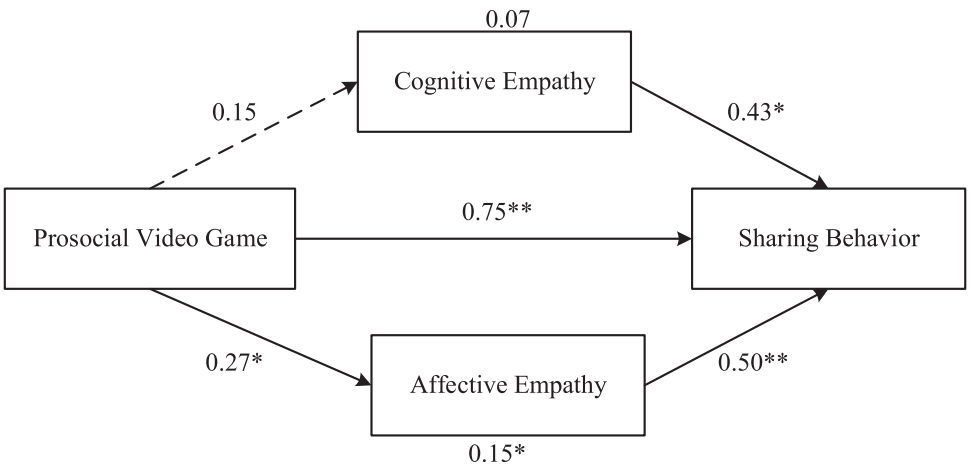

Fig. 3. *Mediation Model of Prosocial Video Games on Sharing Behavior through Affective Empathy and Cognitive Empathy. Note.* 0 = playing neutral game, 1 = playing PVG. Unstandardized path coefficients; Solid lines represent a significant path; dashed lines represent a non-significant path. * $p < .05$; * * $p < .01$.

mediating effect of AE was also found to be significant for girls (ACME = 0.29, 95% CI [0.12, 0.67], $p = .000$) with a proportion of 20.50%, but not for boys (ACME = 0.06, 95% CI [−10.90, 0.01], $p = .600$). In general, the mediating effect of empathy differs between girls and boys, with the effect being more pronounced in girls than in boys. This is similar to AE in that the mediating effect is only

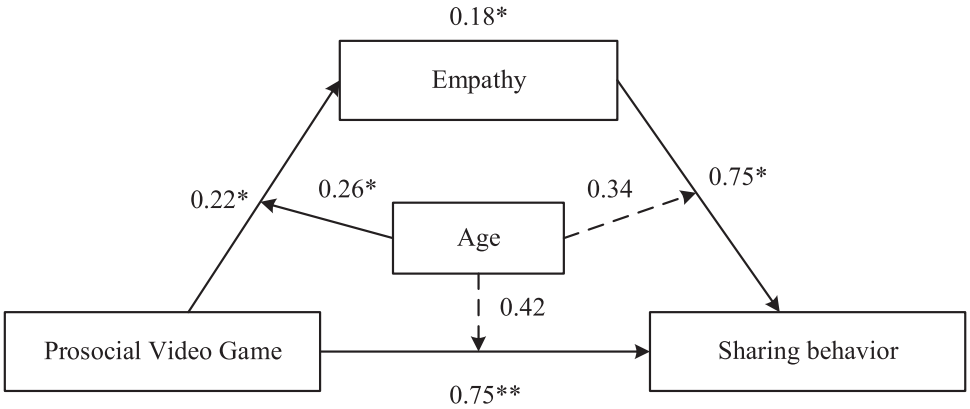

Fig. 4. *Moderated Mediation Model with Age as a Moderator. Note.* 0 = playing neutral games, 1 = playing PVG. Unstandardized path coefficients; Solid lines represent a significant path; dashed lines represent a non-significant path. * $p < .05$; * * $p < .01$; * ** $p < 0.001$.

present for girls.

4. Discussion

Taken together, in the present work, playing PVGs led to an increase in children's sharing behavior. Empathy, especially AE, mediated the association between playing PVGs and sharing behavior. Furthermore, the mediation effect was only found for older children and girls.

Consistent with Hypothesis 1, the present work found that brief exposure to a PVG can lead to more sharing behavior than a neutral video game, extending previous findings on prosocial behavior to younger children (Greitemeyer & Osswald, 2011). Why does playing the PVG improve a children's sharing behavior? Social learning theory could provide a plausible explanation for the increased sharing behavior as a result of playing a PVG. This theory suggests that individuals' behavior is influenced by their environment and the behavior of others around them (Bandura, 1969a; Grusec, 2020). In the current study, players were required to help game characters to clean up litter for the community, which could create a positive environment for learning prosocial behaviors. This cooperative and interactive environment might positively influence the player's behavior towards others in a positive way in real-life situations as they seek to replicate the positive outcomes they observe in others. Therefore, they were more willing to share what they valued with others after playing the PVG. This finding provides a valuable opportunity for teachers and parents to cultivate prosocial behaviors and values through game-based activities and discussions, both in the classroom and at home. For instance, teachers can integrate relevant PVGs into their curricula to promote peer interaction and collaboration. Similarly, as part of a balanced approach to leisure and entertainment at home, parents can facilitate discussions about the prosocial themes of games and help children apply these principles to real-life situations.

Empathy partially mediated the effects of PVG on sharing behavior, but the mediating effect was only for AE, not CE. Based on the social learning theory, there is a mediating process between behavioral stimuli from physical space, television, movies and an individual's performance in real life (Bandura, 1969a). In the case of PVGs, where players engage in behaviors that benefit others, empathy plays a crucial role in mediating the link between social learning and behavioral performance (Gentile et al., 2009; Gentile et al., 2014; Zhang & Wang, 2019). By observing game characters engaging in prosocial behavior, individuals could gain insight into the positive impact of such actions on others. This could lead to the development of empathy as individuals come to appreciate the emotional experience of others and the value of helping others. In turn, increased empathy could motivate individuals to subsequently engage in more prosocial behavior - to share more with others. Furthermore, the mediating effect was only for AE and not CE, replicating another study which found that the affective processing pathway was the most influential in predicting the short-term effects of prosocial media on children (Barlett & Anderson, 2012). This can be explained by the developmental characteristics of cognitive and emotional development in early childhood. At this stage, emotional development often precedes cognitive ones, so children may be better able to understand and experience the emotions of others than to have a thorough cognitive understanding of their behavior (Bensalah et al., 2016; Stietz et al., 2019). As a result, playing the PVG may be more emotionally resonant for children in the short term, and the effects of CE can be tested in older children in a future study. The present study confirmed the viewpoints of social learning theory. Notably, it revealed a significant mediating role of AE in the relationship between observation and young children's sharing behavior. This finding highlights the critical importance of emotional factors in the social learning process and sheds light on their potential influence on the acquisition and manifestation of prosocial behavior. Therefore, it may be effective for educators to use a PVG that promotes emotional resonance in kindergarten to cultivate children's sharing behavior.

The present work also found that the mediation effect was moderated by age and gender, which is consistent with the social learning theory that individual factors may influence an individual's ability to learn from video games (Anderson & Bushman, 2002). Specifically, the mediating effect of empathy, especially AE, was only present for 5- and 6-year-olds, but not for 4-year-olds. This can be explained by the higher level of empathy of older children (Lennon et al., 1986; Marcus et al., 1979). Concretely, as children get older,

their emotional abilities develop and mature, making them better able to understand the feelings and needs of others and to have a corresponding emotional response (Sanchis-Sanchis et al., 2020). In this case, older children may be more likely to show empathic responses during the sharing test after playing PVGs. It is worth noting that the mediating effect of AE was only significant for older, but not for younger children, which may be attributed to the fact that the affective component of empathy is more easily activated in older children than in younger ones. Studies suggest that as children grow older, there is a shift towards the involvement of additional medial parts of the anterior frontal lobe in AE processing, but not in CE (Brink et al., 2011). In other words, the medial part of the anterior frontal lobe becomes more involved in emotional processing as children grow older, whereas it is difficult for children as young as 4 years old to utilize this region effectively.

Similarly, the study found that the mediating effect of playing PVGs was only observed in girls, not boys. A plausible explanation for this finding is the tendency for girls to be more empathetic in response to others' adversity, as supported by previous research (Benenson et al., 2021; Busching & Krahé, 2020). This gender disparity in empathy may be attributed to the traditional caregiving role often associated with females across mammalian species (Decety, 2010). In addition, neuroscience research has shown that girls have larger grey matter volumes, which are strongly associated with empathy, compared to boys (Cheng et al., 2009; Takeuchi et al., 2014). In this study, after observing prosocial behavior in video games, girls may be more likely than boys to experience increased empathy and to show a significant empathic response in subsequent sharing tasks. Furthermore, the mediating effect of AE was significant for girls as opposed to boys. Previous research has shown that video games with more action-oriented content tend to be marketed to boys, while those with more educational and affiliative content are marketed to girls (MacGowan & Schmidt, 2021). As a result, girls may have more experience with the affiliative influence of PVGs, thereby increasing their likelihood of empathizing with others during sharing activities. It is important to note, however, that while the present study did not find statistically significant evidence to support the cultivation of empathy in boys using PVGs, this does not mean that boys are devoid of potential benefits from engaging with such games. Many video games include elements of strategic thinking, problem solving and decision making, which are critical in developing cognitive skills such as critical thinking, spatial awareness and hand-eye coordination, areas in which boys tend to be strong (Venkatesh et al., 2000; Voyer et al., 1995). To consolidate and extend these findings, future research could explore these hypotheses within a larger pool of participants. Furthermore, there seemed to be a marginally significant moderating effect of gender on the associations between video game playing and empathy, indicating a possible trend towards significance. To further explore this moderating effect, a study with a larger sample size would be valuable.

The current study used a between-subjects design and randomized participants, which could provide evidence for the association between playing PVGs and sharing behavior, as well as the mediating and moderating mechanisms behind this association. However, this study has several limitations. First, the current study measured children's trait empathy and future research should consider measuring state empathy as well to determine whether it also has a mediating effect similar to trait empathy. State empathy, is a more transient and context-dependent type of empathy that reflects an empathic response in a particular situation or context (Shen, 2010). Although some studies have found that prosocial media use predicts increases in trait empathy in the short term (Greitemeyer et al., 2010), it is possible that state empathy plays a more significant mediating role between playing PVG and sharing behavior. Therefore, it is important to replicate the current findings using measures of state empathy. In addition, a multi-informant or triangulated approach may be used to test empathy to overcome the limitations of self-report in younger children in future studies. Second, pre-tests of sharing behavior and empathy are needed to better determine the effects of PVGs and neutral video games in future research. Without a pre-test, we do not know whether playing PVGs increased empathy and sharing compared to baseline, or whether playing neutral video games decreased empathy and sharing compared to baseline. In order to get a more concrete picture of the development and change in children's sharing behavior and empathy, the pretest of sharing behavior and empathy should be administered in the future. Third, future studies should consider measuring the long-term effects of PVG. This study looked at the immediate effects of PVG, it is not clear how long these effects last. Therefore, there is also a need to provide evidence of the long-term effects of PVG so that they can be considered in the long-term education of children. The study's fourth limitation is the absence of information regarding participants' prior experience with video games and other relevant demographic variables. This lack of data prevents us from discerning whether children's previous gaming experience has an impact on their performance in gaming tasks, as well as their subsequent performance in sharing tests. Consequently, to gain a more comprehensive understanding of the influence of video game play on children's sharing behaviors, it is helpful to include a more extensive range of demographic information in future studies, with a particular focus on capturing participants' prior experience with video games. Finally, the mediating effect of CE was not significant in this study. Older children (who are more cognitively competent) as participants can be considered in the future to investigate the effect of PVG on their sharing behavior.

This study revealed that brief exposure to PVGs in preschool children might increase the accessibility of sharing behavior compared to neutral video games, suggesting that PVGs might be a useful tool for educators to promote sharing behavior in children. Empathy, especially AE, was found to mediate the association between playing PVGs and sharing behavior. This suggests that empathy training programs, especially those that promote AE, may be an effective means of promoting sharing behavior in children. The mediating effect was mainly observed for girls and older children. Girls and older children may therefore be the target population for sharing behavior education programs that aim to increase empathy in playing PVGs. However, promoting sharing behavior in younger children and boys is equally important and should be given attention in future research.

Institutional Review Board Statement

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee, Faculty of Education, Northeast Normal University (20200601.02 and date of 1 June 2020).

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